Parametric Survival Analysis

Colon Cancer Dataset

Survival Analysis Group Project

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Data Summary — Colon Cancer Dataset

Table 1: Sample Summary

Statistic	Value
Total Patients	929
Events (Deaths)	452 (48.7%)
Censored	477 (51.3%)
Min Follow-up	23 days
Max Follow-up	3,329 days
Mean Follow-up	1,670 days
Median Follow-up	1,976 days

Missing data: `nodes` had 18 NAs (<2%) — imputed via **LOCF**.

Table 2: Descriptive Statistics of Key Variables (Pre-Imputation)

Variable	Min	Max	Mean	Med.
Time (days)	23	3329	1670.0	1976
Status	0	1	0.49	0
Age (years)	18	85	59.75	61
Sex	0	1	0.52	1
Nodes	0	33	3.66	2

Status: 0=censored, 1=death **Sex:** 0=female, 1=male

Model Fitting — All Parametric Models

Table 3: Parametric Model Comparison

Model	MLE Parameters	LogLik	AIC	BIC
Gen.Gamma	μ=7.545; σ=1.575; Q= − 0.481	−4103.5	**8212.9**	8227.4
Log-Normal	μ_ℓ=7.777; σ_ℓ=1.432	−4106.6	8217.1	8226.8
Log-Logistic	shape= 1.202; scale= 2309.7	−4116.4	8236.8	8246.4
Gompertz	shape= $-3{\cdot}10^{-4}$; rate= $4{\cdot}10^{-4}$	−4123.6	8251.2	8260.8
Exponential	rate= $3{\cdot}10^{-4}$	−4131.7	8265.4	8270.3
Gamma	shape= 1.050; rate= $3{\cdot}10^{-4}$	−4131.3	8266.7	8276.4
Weibull	shape= 0.999; scale= 3433.9	−4131.7	8267.4	8277.1

Fitted via MLE using `flexsurvreg()`.

Best Model Selection — Generalised Gamma

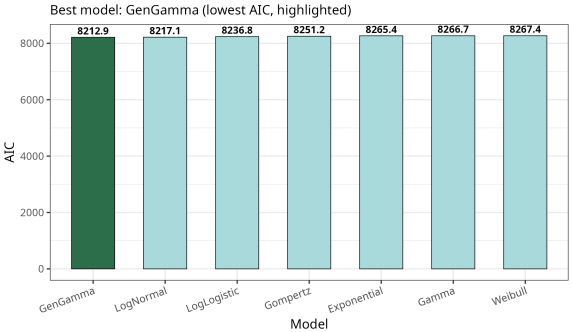

Figure 1: AIC values across all seven parametric models.

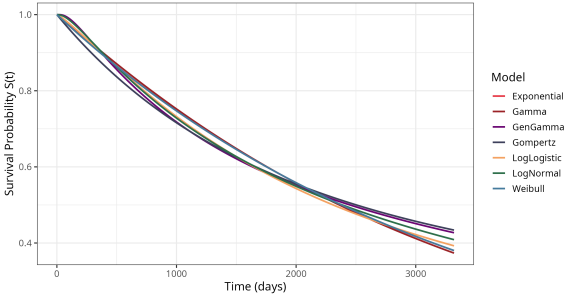

Figure 2: Fitted survival curves for all models. Divergence after $t > 2000$ days; Gen.

Justification: Lowest AIC (8,212.9), highest log-likelihood (-4103.5), best visual fit at long follow-up. ΔAIC vs. Log-Normal $= 4.2$.

Life Functions — Generalised Gamma Model

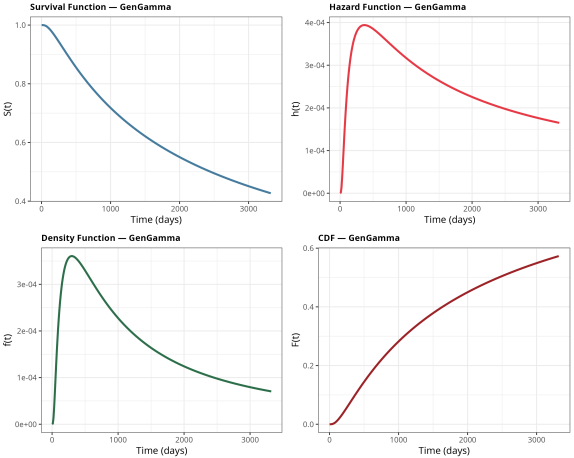

Figure 3: Life functions.

Interpretation — Generalised Gamma Model

Table 5: MLE Parameter Estimates with SE and 95% Confidence Intervals

Param.	Est.	SE	95% CI
μ	7.545	0.117	[7.317, 7.774]
σ	1.575	0.073	[1.438, 1.725]
Q	-0.481	0.191	$[-0.855, -0.106]$

- μ=7.545: median $\approx e^{7.545} \approx 1{,}900$ days
- σ=1.575: spread of survival
- Q=-0.481<0: non-monotone hazard

Life function interpretation:

- **S(t):** 90% survive at 1 year; ≈55% at 5 years
- **h(t):** Risk peaks near day 300 then gradually declines
- **f(t):** Most deaths near day 300 post-surgery
- **F(t):** 50% die within ≈2,500 days (∼6.8 yrs)

Mean survival	2,130 days (5.84 yrs)
Std deviation	1,221 days
Variance	1,491,722 days^2

Conclusions & Clinical Insights

Statistical Conclusions

- **Generalised Gamma** is the best model (AIC = 8,212.9) among 7 candidates
- Flexible hazard shape outperforms Weibull, Exponential and Gamma
- $Q<0$ confirms a **non-monotone hazard** not captured by standard models
- High SD (1,221 days) reflects substantial **patient heterogeneity**

Clinical Implications

- **Critical window:** highest death risk in the first ∼300 days — intensive post-surgical follow-up is essential
- **Mean survival 5.8 years** is a realistic planning benchmark
- ∼45% survive beyond 3,000 days — a favourable long-term subgroup exists
- `nodes` count should be explored in future covariate models

The non-monotone hazard identified by the Generalized Gamma model supports a biphasic clinical strategy: intensive early monitoring during the first year post-surgery, followed by structured long-term survivorship care thereafter